

Industrial Electronics

Engineering Technology

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Short Title:	Industrial Electronics	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Electronics	
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Credits	5	Level: Intermediate

Description of Module / Aims

The aim of this module is to provide the student with an understanding of power electronic circuits, equipment and related calculations. A focus is kept on the modern application of these devices to renewable, industrial and automotive applications. In addition, the student will build and test a selection of the circuits in laboratories.

Programmes

	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	2 / 4 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	3 / 6 / M
	BEng (Hons) (WD_ETRIC_B)	2 / 4 / M
ELTR-0031	BEng in Electrical Engineering (WD_EELCT_D)	3 / 5 / M

Indicative Content

- Thyristors, Triacs, IGBT, Mosfet, Transistor Power Switches.
- DC Motor Control.
- DC-DC converters
- Switching Power Supplies
- AC-DC rectification
- Inverters, industrial and automotive applications.
- Energy recovery in DC and AC motor control circuits.
- Battery Technology
- Battery Chargers
- UPS
- Typical Mini Project; brushless dc motor controller

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the operation, application, control and function of typical power switches.
2. Define the topology, application, control, perform calculations and describe the function of typical dc motor drives, ac rectifiers, ac motor drives, UPS systems and brushless dc motor controllers.
3. Define the topology, application, control, perform calculations and describe the function of typical dc-dc, switch mode power supplies typical battery technology.
4. Build, test, analyse and report on the practical power electronic circuits in accordance with the module submission schedule.

Learning and Teaching Methods

- The course will consist of lectures, laboratories, mini project and formative assessments.

- The emphasis on course delivery will be to engage the student in the learning of the material, through the integration of the course materials, with the real case studies in laboratories and mini project.
- The formative assignments will be used to provide an opportunity for the student to engage with the material, before it is applied, to maximise the learning opportunities for the student.
- Self-directed learning, use of moodle, and recommended software.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3
Continuous Assessment	40%	
<i>In-Class Assessment</i>	10%	1,2,3
<i>Project</i>	10%	4
<i>Practical</i>	20%	4

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Moham, N. *_First Court on Power Electronics & Drives_*. UK: Mnpere, 2003.
- Rashid, M.H. *_Power Electronics Handbook_*. UK: Academic Press, 2001.

Requested Resources

- ENGINEERING LAB: Electronics